

Sales Data Analysis Using SQL

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Project: Sales Analysis Using SQL

Tools Used: MySQL, Tableau, Google Docs

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1. Introduction

Dataset

The dataset contains sales transactions from April that I found on Kaggle and includes information such as order ID, product, quantity ordered, price of each item, order date, and purchase address. This data can be used to look at customer purchases, product performance, revenue, and sales trends.

Purpose

The purpose of this project is to analyze April sales data using SQL and find useful business insights. The analysis looks at sales performance, customer purchases, product popularity, and revenue trends. SQL was used to answer questions such as which products generated the most revenue, what time customers placed the most orders, which city had the highest sales, and which products were commonly purchased together.

SQL Skills Demonstrated:

Aggregate functions, GROUP BY, ORDER BY, date functions, string functions, subqueries, self joins, and CASE statements.

2. Business Questions & Analysis

2.1 What Was the Total Revenue Generated During April?

SQL Approach:

Multiplied the quantity ordered by the price of each item and then summed the revenue across all transactions in April to find the total revenue.

Result:

The total revenue generated during April was **\$3,396,059.11**.

Business Insight:

Total revenue gives an overall measure of sales performance for the month and provides a baseline for comparing performance in different areas.

2.2 What were the top 5 products by revenue?

SQL Approach:

Multiplied the quantity ordered by the price of each item to find revenue. Grouped the data by product, sorted the revenue from highest to lowest, and limited the results to the top five products.

Result:

1. MacBook Pro Laptop — **\$773,500.00**
2. iPhone — **\$486,500.00**
3. ThinkPad Laptop — **\$391,996.08**
4. Google Phone — **\$348,600.00**
5. 34in Ultrawide Monitor — **\$248,133.47**

Business Insight:

The five products that generated the most revenue were mostly high-priced electronics, showing the importance of higher-priced products to overall revenue. This information can be used to plan marketing and promotions around products that generate the most revenue.

2.3 What were the top 5 products by units sold?

SQL Approach:

Calculated the total number of units sold by adding the quantity ordered for each product. Grouped the data by product, sorted the total units sold from highest to lowest, and limited the results to the top five products.

Result:

1. AAA Batteries (4-pack), **2,936 units**
2. AA Batteries (4-pack), **2,822 units**
3. Lightning Charging Cable, **2,371 units**
4. USB-C Charging Cable, **2,270 units**
5. Wired Headphones, **2,066 units**

Business Insight:

Lower-priced products sold the most units, while higher-priced products generated the most revenue. This helps identify which products have the highest demand and shows that selling the most units does not necessarily mean generating the most revenue.

2.4 How many unique orders were placed?

SQL Approach:

Used COUNT(DISTINCT) on the order ID to find the total number of unique orders.

Result:

A total of **17,537 unique orders** were placed during April.

Business Insight:

The number of unique orders gives a measure of order volume during the month, which can be used as a baseline for comparing performance across different time periods and identifying trends in customer purchases.

2.5 What hour of the day had the most orders?

SQL Approach:

Extracted the hour from the order date and counted the number of unique orders placed during each hour. The results were then sorted from highest to lowest to identify the busiest hour of the day.

Result:

The highest number of orders occurred at **7 PM** , with **1,225 unique orders**.

Business Insight:

We can identify the hours when customers are most active and use this information to plan the best times for marketing and promotions.

2.6 Which day generated the most revenue?

SQL Approach:

Multiplied the quantity ordered by the price of each item and added the results together to find total revenue. Grouped the data by date and sorted the revenue from highest to lowest to find the day that generated the most revenue.

Result:

The highest revenue-generating day was **April 24, 2019**, with total revenue of **\$136,177.32**.

Business Insight:

Finding the day that generated the most revenue can help identify patterns such as higher order volume, more expensive products being purchased, or other factors that contributed to stronger sales performance. These patterns could help the business improve future sales.

2.7 Which city generated the highest revenue?

SQL Approach:

Used the purchase address to extract the city using TRIM() and SUBSTRING_INDEX(). Multiplied

the quantity ordered by the price of each item to find total revenue. Grouped the data by city and sorted the total revenue from highest to lowest.

Result:

San Francisco generated the highest revenue, with a total of **\$817,074.77**.

Business Insight:

Finding the city that generated the most revenue can help determine which areas the business should target more. Since San Francisco generated the highest revenue, the business could look at what is driving its strong performance and use those insights when planning advertising, inventory, and marketing strategies in other locations.

2.8 What was the average revenue per order?

SQL Approach:

Multiplied the quantity ordered by the price of each item to find total revenue, then grouped the results by order ID to find the total revenue for each order. Used a subquery to calculate the average revenue across all individual order totals.

Result:

The average revenue per order was **\$193.65**.

Business Insight:

Average revenue per order gives a measure of how much customers typically spend per order and provides a baseline for comparing customer spending across different areas or time periods.

2.9 Which products were most frequently purchased together?

SQL Approach:

Used a self join to compare products within the same order by matching rows with the same order ID. The matching product pairs were then grouped and counted to find how frequently each combination was purchased together.

Result:

1. Google Phone + USB-C Charging Cable — **106**
2. iPhone + Lightning Charging Cable — **106**
3. iPhone + Wired Headphones — **45**
4. Google Phone + Wired Headphones — **42**
5. USB-C Charging Cable + Vareebadd Phone — **38**
6. Apple AirPods Headphones + iPhone — **37**
7. Bose SoundSport Headphones + Google Phone — **24**
8. USB-C Charging Cable + Wired Headphones — **16**
9. Vareebadd Phone + Wired Headphones — **15**
10. AA Batteries (4-pack) + Lightning Charging Cable — **14**

Business Insight:

The results show a strong relationship between smartphones and compatible accessories. The top four product combinations included a phone paired with either a charging cable or wired headphones. Knowing this, the business could create bundle deals or promotions that encourage customers to purchase related accessories with their phones.

2.10 How many Small, Medium, and Large orders were placed?

SQL Approach:

Calculated the total revenue for each order by grouping the results by order ID. Used a CASE statement to categorize each order as Small (under 100), Medium (100 to 300), or Large (300 or more). Then counted the number of orders in each category.

Result:

1. Small — **10,801 orders**
2. Large — **4,136 orders**
3. Medium — **2,600 orders**

Business Insight:

Small orders made up the majority of purchases during April, showing that customers were more likely to place lower-value orders. Knowing the different order sizes can help the business better understand customer spending and find ways to encourage customers with smaller orders to spend more.

3. Overall Findings

Higher-priced electronics such as the MacBook Pro Laptop, iPhone, ThinkPad Laptop, and Google Phone generated the most revenue, while lower-priced products such as batteries, charging cables, and headphones sold the most units. This shows that the products that sell the most units are not always the products that generate the most revenue. There was also a strong relationship between smartphones and their accessories. Phones were commonly purchased with charging cables and headphones. This could be useful for creating bundle deals or promotions that motivate customers to buy related products together. San Francisco generated the most revenue out of all the cities analyzed. The number of orders also changed depending on the time of day, showing that customers were more active during certain hours. This information could help the business decide where and when to focus marketing and promotions. The average revenue per order was \$193.65, and most orders were classified as Small, with 10,801 orders under \$100. There were also 4,136 large orders of \$300 or more. Looking at order sizes helps show how much customers typically spend and how customer spending can vary.

4. Conclusion

This project used SQL to analyze April sales data and answer business questions about revenue, products, customer purchases, orders, and sales trends by location. The analysis showed that higher-priced electronics generated the most revenue, while lower-priced products sold the most units. It also showed that phones were commonly purchased with related accessories and that sales performance differed depending on location and time. Throughout the project, I used different SQL techniques to answer these questions and find patterns in the data. These included aggregate functions, GROUP BY, date and string functions, subqueries, self joins, and CASE statements.